F70LB Life Insurance Mathematics B

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14 Pension Funds

A pension is an income paid to someone from the time they cease employment for the rest of their life.

The pension can be paid by an insurance company or the worker's employer, or the government. This chapter provides a very brief introduction to the concept of pension funds.

More advanced topics about pension funds are covered in the MSc Actuarial Management and in the professional examinations.

14.1 An Introduction to Pension Funds

Pensions are important because:

- They enable people to retire at a reasonable age.
- Pension funds account for a significant proportion of the stock market
- Pension benefits may represent an important part of an employee's remuneration package
- About 30% of IFoA actuaries work in pensions compared to 37% in life insurance and 17% in general insurance.

Who pays pensions?

- An individual and/or their employer may pay premiums to an insurance company, which will then provide the pension in retirement. For example, personal pensions.
- An employer may run an occupational pension scheme (OPS) into which contributions are paid while an employee is working, and which then pays the employee's pension in retirement.
- The government may pay a state pension funded by taxes or contributions.

Most pensions actuaries work with OPSs. The traditional form of OPS in the UK offers a pension related to the retiree's salary. These are a type of Defined Benefit (DB) Scheme.
The other main type of pension scheme is called a Defined Contribution (DC) Scheme.

14.1.1 Defined Benefit Schemes

The link to earnings may be:

- A proportion of the retiree's salary on the retirement date.
- A proportion of the retiree's average salary over some period prior to their retirement date (e.g. over the last 3 years).
- A proportion of the retiree's average salary over some longer period (e.g. their whole working lifetime with the company).

Pensions in proportion to earnings allow the pensioner to maintain a standard of living comparable with that they enjoyed during their working life.

Also, since contributions paid into OPSs are usually a percentage of salary, those with higher salaries will have made larger contributions.

The pension is always less than 100% of the earnings on which it is based; it is expected that living expenses fall after retirement (mortgage paid off, children grown up, no need to travel to work etc.).

The pension also depends on the length of service with the sponsoring employer; naturally, long service leads to a bigger pension than brief service.

Thus the pension is often expressed as

$$\frac{n}{N} \times S$$

where

- S is salary. This could be final salary at retirement or average salary.
- n is the number of years worked. This may be capped at some maximum M, for example $M = 40$.
- $\frac{1}{N}$ is called the "accrual rate"

N is often 60 or 80 (called '60ths schemes' and '80ths schemes').

Thus a maximum expected working lifetime of 40 years with the same company would earn a pension of 1/2 to 2/3 of final (or average) salary.

14.1.2 Defined Contribution Schemes

In contrast to a DB scheme, an OPS may instead fix the level of contributions as a percentage of salary, and pay these into an investment fund.

On retirement the pensioner receives their share of the investment fund and can use it to buy an annuity from an insurance company, or they can choose to draw down on the fund (like a savings account) over time.

Thus their pension depends on market conditions before retirement (how successfully the fund has been invested) and at retirement (what annuity rates insurers are then offering).

If they choose not to buy an annuity at retirement, then the pensioner accepts investment risk and longevity risk instead of having a guaranteed income.

Either way, the individual's DC pension as a proportion of their final or average salary could be extremely unpredictable.

Such schemes are called Defined Contribution (DC) schemes or (old-fashioned) Money Purchase Schemes.

14.1.3 Risk and pensions

In the UK, most private-sector DB schemes are now closed to new entrants or to all members. This means that the schemes still provide the benefits that members built up before the scheme closed, but members can't build up any further benefits. Workers in the UK public sector, like teachers and police, still have access to DB schemes.

A very important difference between DB and DC schemes is who bears the risk. Differences in risk are one of the main reasons for the move away from DB.

In a DB scheme the provider (usually an employer) takes most of the risk. They have promised to provide a pension according to some fixed formula, but they cannot know in advance what the pension will cost (or even what the rate of pension will be). The cost will depend on future longevity, investment returns and other factors. So in a DB scheme, the employer has the risk of paying higher or lower contributions than they expected.

In a DB scheme, a risk for the member is that the employer is unable to fulfil their promise to pay the full level of pension. In the UK, members are partially protected against that risk through a body called the Pension Protection Fund.

In a DC scheme, the risk falls mostly on the scheme member because contributions are fixed (or can be chosen by the member), but the final pension is unpredictable.

14.1.4 Receiving Benefits - UK DB schemes

Many people like to receive a lump sum on retirement, so many UK DB schemes allow a retiring member to 'commute' part of their pension entitlement into a lump sum. For example, the scheme might give the member £12 of immediate cash lump sum in exchange for each £1 of pension they give up. This is like the reverse of buying an annuity.

DB schemes will specify a normal retirement age (NRA), or equivalent, which is the earliest age at which pension scheme members can start taking their pension. It is often 60, 65 or State Pension Age. (State Pension Age in the UK depends on when you were born, but is higher than age 65.)
Most OPSs recognize that retirement at NRA is not the only way in which a member of an OPS may leave, hence benefits may also be payable upon other events. Examples are:

- Normal retirement at NRA (as described above)
- Early retirement by choice before NRA.
- Early retirement (before NRA) because of ill-health.
- Death in service, i.e. death while employed.
- Withdrawal (leaving the employer but not yet starting a pension, usually due to not yet being old enough).
- "Late" retirement: This means any retirement after NRA.

In each case, the benefits payable will be determined by the scheme rules (or overriding legislation).

Early retirement (by choice) usually means a reduced pension entitlement to reflect the fact that it will be paid for longer. The earliest age this is possible is typically at least 55.

Ill-health retirement means that the member is below NRA and meets a definition in the scheme rules to qualify for a special ill-health pension. This is a lot like an Income Protection Insurance benefit. For example, if the member is permanently too ill to do their own job they might qualify for an ill-health pension. This is a pension that starts early, typically with no reduction (unlike early retirement by choice) and may be enhanced (increased to a higher amount).

On late retirement, the pension might be increased but often it is not.

Death usually results in a dependant's pension (like a reversionary annuity) being paid if the member has a spouse or other eligible dependant. If death occurs in service, then a lump sum will often also be paid and the dependants' pension might be enhanced (increased to a higher amount) as well as being paid immediately (potentially before NRA).

Withdrawal (moving to another employer) results in the entitlement earned to date being frozen until retirement, or the scheme assets and liabilities in respect of the ex-employee being transferred to the OPS of the new employer or another pension arrangement.

14.2 Valuation of the DB Liabilities

14.2.1 Objectives

The objectives in valuing the liabilities may be:

- To calculate the amount of money we need now to afford the benefits we promised (i.e. like a policy value, but normally with no allowance for future income).
- To calculate the amount of money to be paid into the fund in future to afford the benefits promised (i.e. like premiums). Such payments are called contributions and are usually expressed as a percentage of salary.

14.2.2 Decrements

An 'active' member of a DB scheme is one still in employment. We consider four ways in which a member may cease to be active and cause benefits to be paid:

- Age (i.e. normal) retirement
- Ill-health retirement
- Death
- Withdrawal.

These events may be represented by a multiple-decrement model, which is a form of multiple-state model, see Figure 1.

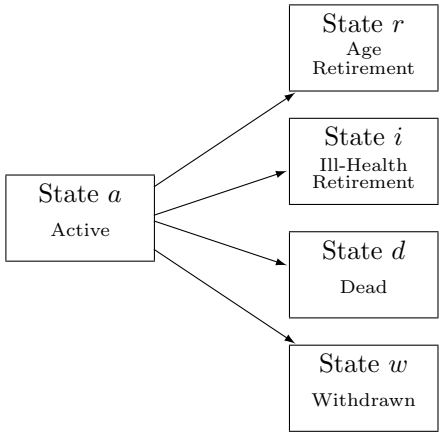


Figure 1: A Multiple Decrement Model for a Pension Scheme.

14.2.3 The Service Table

Based on this model, we define a Service Table, which is exactly analagous to the ordinary life table in the alive-dead model.
Define

- l_α to be the radix of the service table, where α is the lowest age at which members may join the scheme.
- l_x to be the expected number of the l_α 'joiners' still active at age $x > \alpha$, and
- define r_x, i_x, d_x and w_x to be the expected number of the l_x who will retire normally, retire in ill-health, die and withdraw, respectively, before age $x + 1$

Then:

$$l_{x+1} = l_x - r_x - i_x - d_x - w_x$$

and, in multiple-state notation, the probability that a member active at age x is active at age $x + t$ is:

$${}_tp_x^{aa} = {}_tp_x^{\overline{aa}} = \frac{l_{x+t}}{l_x}.$$

Similarly, the probabilities that a member active at age x should suffer any one of the decrements before age $x + 1$ are:

$$\begin{aligned} {}_1p_x^{ar} &= \frac{r_x}{l_x} \\ {}_1p_x^{ai} &= \frac{i_x}{l_x} \\ {}_1p_x^{ad} &= \frac{d_x}{l_x} \\ {}_1p_x^{aw} &= \frac{w_x}{l_x}. \end{aligned}$$

More generally, the probabilities that a member active at age x should suffer any one of the decrements between ages $x + t$ and $x + t + 1$ are:

$$\begin{aligned} {}_tp_x^{aa} \times {}_1p_{x+t}^{ar} &= \frac{r_{x+t}}{l_x} \\ {}_tp_x^{aa} \times {}_1p_{x+t}^{ai} &= \frac{i_{x+t}}{l_x} \\ {}_tp_x^{aa} \times {}_1p_{x+t}^{ad} &= \frac{d_{x+t}}{l_x} \\ {}_tp_x^{aa} \times {}_1p_{x+t}^{aw} &= \frac{w_{x+t}}{l_x}. \end{aligned}$$

Examples: p142 of the Orange Tables shows an example service table. In real situations, every scheme is different and has its own table.
Considering the example service table in the Yellow Tables, we see the following:

- Withdrawal: Probability is initially high, but decreases with age. (Sometimes you will also see temporary initial selection in withdrawal rates.)
- Death: Probability is initially low, but increases with age.
- Ill-health: Probability increases with age, but is very low, rises at age 60.

- Age retirement:
  - Normal retirement age (NRA) = 65
  - Early retirement: Allowed from 60 to 65.

14.2.4 The Salary Scale

It is expected that salary will increase over time due to:

1. inflation
2. promotion
3. experience.

Hence let s_α be an arbitrary positive number and define s_x for $x > \alpha$ by:

$$\frac{s_{x+t}}{s_x} = E\left(\frac{\text{Salary Earned } x+t \to x+t+1}{\text{Salary Earned } x \to x+1}\right)$$

assuming that the employee is in service for the entirety of each period considered.
We call the function s_x a 'salary scale'.

Example: Suppose the salary earned between ages 18 and 19 is £18,000. Based on the salary scale in the Yellow Tables, calculate the expected salary earned between: (i) ages 19 and 20; and (ii) ages 40 and 41.
Solution:

$$\begin{aligned} s_{18} &= 1.927 \\ s_{19} &= 2.204 \\ s_{40} &= 7.814. \end{aligned}$$

$$\begin{aligned} \text{Expected salary between 19 and 20} &= 18K \times \frac{2.204}{1.927} \\ &= 20,587 \\ \text{Expected salary between 40 and 41} &= 18K \times \frac{7.814}{1.927} \\ &= 72,990. \end{aligned}$$

14.2.5 The Relationship Between Salary Earned and Salary Rate

- The salary rate at age x is the instantaneous rate of payment at that age.
- The salary earned is the amount of salary paid over a time interval.

Thus, the salary earned between ages x and $x+t$ is:

$$\int_0^t (\text{Salary rate at age } x+s)ds.$$

Sometimes we assume that the salary rate increases continuously through the year.
Then the salary earned between ages x and $x+1$ is approximately equal to the the salary rate at age $x+1/2$. This is exact if the rate of salary increases linearly between ages x and $x+1$.

Sometimes it is more realistic to assume that salaries increase once a year, on a birthday or on a fixed calendar date.

Example: Suppose the salary rate at age 30 is $£20,000$ and that salary rate increases by $£4,000$ at age 30.75. Calculate the salary earned between ages 30 and 31.

Solution:

$$\text{Salary earned} = \frac{3}{4} \times 20K + \frac{1}{4} \times 24K = 21K.$$

14.2.6 Final Pensionable Salary (FPS)

The scheme rules will specify the pension in terms of Final Pensionable Salary (FPS). FPS can be any function of the current or past salaries. A common definition is the average salary earned over the last n years of service.